

Activity: Exploring EV Evacuation Routing and Mobile Charging Station Deployment

1. Summary

In this activity, participants use the EV evacuation optimization code to study how mobile charging-station availability, initial EV driving range, and evacuation flow affect feasible routing, charging decisions, evacuation time, and MCS deployment. Participants first run a baseline case, then change selected parameters and compare the modified results with the baseline.

The workflow mirrors a baseline-versus-modified-scenario exercise: record assumptions, predict the effect of a change, run the code, record the terminal summary, and interpret the differences.

2. Learning objectives

- Explain how the total number of available MCSs can affect routing feasibility and charging support.
- Explain how initial EV driving range can change the need for FCS/MCS charging along an evacuation route.
- Explain how OD-pair evacuation flow can affect road and charging-capacity constraints.
- Compare baseline and modified optimization results using charging time, travel time, total evacuation time, and MCS deployment.
- Identify parameter combinations that make the optimization difficult to solve or infeasible.

3. Preparation and materials

- A computer with Python and the packages required by the repository.
- The optimization script and its supporting solver/data files.
- A working baseline case that produces a feasible solution.
- This participant worksheet for recording baseline and modified results.

Before the activity, the instructor should run the baseline case once and verify that the terminal summary and route plots are produced. Participants should modify only the designated parameters shown in this worksheet.

4. Baseline scenario

Use the current default values in the optimization code as the baseline.

Parameter	Baseline value	Where it is changed
Total number of available MCSs	25	num MCS
Initial flow - each of 7 OD pairs	60 Veh/hr	init_flow
Initial range - each of 7 OD pairs	20 km	init_range
Number of OD pairs	7	start_node_centroid / goal_node_centroid

5. Baseline OD pairs and inputs

OD pair	Start centroid	Goal centroid	Initial flow (veh/hr)	Initial range (km)
OD 1	51736	50623	60	20
OD 2	51731	50626	60	20
OD 3	50668	50606	60	20
OD 4	50662	50611	60	20
OD 5	50659	50614	60	20
OD 6	50677	51851	60	20
OD 7	50637	51874	60	20

6. Suggested agenda

Stage	Time
Instructor introduces the model, baseline case, and editable parameters	10 minutes
Groups predict outcomes and run the baseline	10 minutes
Groups modify one or more assigned parameters and rerun	20 minutes
Groups compare baseline and modified results	15 minutes
Groups report findings	5 minutes

7. Activity instructions

Part A - Predict and run the baseline

Before running the baseline, discuss:

1. How many OD pairs do you expect will require charging?
2. Which input is likely to have the strongest effect on feasibility: total MCS availability, initial range, or OD-pair flow?
3. If the total number of available MCSs is reduced, do you expect travel time, charging time, MCS deployment, or feasibility to change first?

Run the baseline and save or copy the terminal summary. Record the results in the participant worksheet.

Part B - Change selected parameters

Restore the baseline before each independent experiment unless your instructor specifically asks for a combined scenario. Participants may change the total MCS availability, the initial range of one or more OD pairs, and/or the flow of one or more OD pairs.

Example investigation	Parameter change
Limited mobile-charging resources	Reduce num_MCS below the baseline value of 25.
Improved mobile-charging resources	Increase num_MCS above 25.
Low-range OD pair	Reduce init_range for one or more selected OD pairs.
Higher evacuation demand	Increase init_flow for one or more selected OD pairs.
Combined stress case	Reduce range while increasing flow and/or reducing MCS availability.

8. How to change the parameters in the code

Open the main optimization script and locate the baseline OD-pair input block. Change only the values inside `init_flow` and `init_range`; keep the number of entries equal to the number of OD pairs.

Baseline OD-pair inputs

```
67
68 ##### Actual parameters #####
69 start_node_centroid = np.array([51736,51731,50668,50662,50659,50677,50637])
70 goal_node_centroid = np.array([50623,50626,50606,50611,50614,51851,51874])
71 num_od_pair = start_node_centroid.size # change this parameter for multiple od pair
72 init_flow = np.array([60,60,60,60,60,60,60]) # change this parameter for multiple od pair
73 init_range = 2*np.array([[10,10,10,10,10,10,10]]) # change this parameter for multiple od pair
74 #####
```

Figure 1. Baseline OD-pair input block in the optimization code.

Example: to change only OD 1 from 60 EVs and 20 km to 80 EVs and 15 km, edit the first entry in each array:

Baseline	Example modified case
<code>init_flow = [60,60,60,60,60,60,60]</code>	<code>init_flow = [80,60,60,60,60,60,60]</code>
<code>init_range = 2*[[10,10,10,10,10,10,10]]</code>	<code>init_range = [[15,20,20,20,20,20,20]]</code>

Note: `init_range` is the initial driving range in kilometers. In the baseline code, `2*np.array([[10,...]])` produces 20 km for every OD pair. You may either keep the multiplier form or directly enter the desired kilometer values in the array.

To change the total number of MCSs available to the optimization, locate `num_MCS` in the MCS parameter section.

Baseline MCS availability

```
299
300 multiple_MCS_number = 20      # Total number of MCS allowed per site
301
302 MCS_power_kW = 150
303 MCS_range_per_hour = (3.3*1.6)*MCS_power_kW
304
305 MCS_ports = 5
306 MCS_flow_limit = MCS_ports*scaling_factor
307
308 num_MCS = 25  # Total number of MCS available
309 MCS_flow_limit_vector = MCS_flow_limit*np.ones(num_MCS_location)
```

Figure 2. Baseline total MCS availability in the optimization code.

For example, change `num_MCS = 25` to `num_MCS = 15` to study a resource-constrained case. After changing inputs, save the file and rerun the optimization.

Participant Worksheet

Group members: _____

Assigned scenario / question: _____

1. Record your assumptions

Complete the modified-scenario column. Leave all other parameters at their baseline values unless instructed otherwise.

Input	Baseline	Modified scenario
Total number of available MCSs	25	
OD 1 flow: 51736 -> 50623	60	
OD 1 initial range (km)	20	
OD 2 flow: 51731 -> 50626	60	
OD 2 initial range (km)	20	
OD 3 flow: 50668 -> 50606	60	
OD 3 initial range (km)	20	
OD 4 flow: 50662 -> 50611	60	
OD 4 initial range (km)	20	
OD 5 flow: 50659 -> 50614	60	
OD 5 initial range (km)	20	
OD 6 flow: 50677 -> 51851	60	
OD 6 initial range (km)	20	
OD 7 flow: 50637 -> 51874	60	
OD 7 initial range (km)	20	

2. Predict the effects before running the modified case

Output	Predicted change	Reason
Feasibility		
Charging time		
Travel time		
Total evacuation time		
Total deployed MCSs		

3. Record the baseline optimization results

Use the terminal summary generated by the optimization code. Record time in minutes and MCS counts exactly as reported.

OD pair	Charging time (min)	Travel time (min)	Total evacuation (min)	MCS used	Feasible?
OD 1 51736->50623					Yes / No
OD 2 51731->50626					Yes / No
OD 3 50668->50606					Yes / No
OD 4 50662->50611					Yes / No
OD 5 50659->50614					Yes / No
OD 6 50677->51851					Yes / No
OD 7 50637->51874					Yes / No

4. Record the modified-scenario results

OD pair	Charging time (min)	Travel time (min)	Total evacuation (min)	MCS used	Feasible?
OD 1 51736->50623					Yes / No
OD 2 51731->50626					Yes / No
OD 3 50668->50606					Yes / No
OD 4 50662->50611					Yes / No
OD 5 50659->50614					Yes / No
OD 6 50677->51851					Yes / No
OD 7 50637->51874					Yes / No

5. Aggregate comparison

Metric	Baseline	Modified	Main observation
Total initial flow			
Total / available MCSs			
Total deployed MCSs			
Average OD evacuation time (min)			
Maximum OD evacuation time (min)			
Number of OD pairs using MCS charging			

6. Compare the baseline and modified cases

For each question, support your answer with values from the terminal summary or route plots.

1. Which OD pair changed the most in total evacuation time? Why?

2. Did changing initial range mainly affect charging time, route choice, MCS deployment, or several of these?

3. Did increasing flow create a capacity bottleneck on roads, at FCS locations, or at candidate MCS locations?

4. How did changing the total number of available MCSs affect the number actually deployed?

5. Did any modified case reach the solver time limit? If yes, was a feasible solution found before the limit?

6. If no feasible solution was found, which input change appears most responsible?

7. Attach evidence from the run

Paste a screenshot of the baseline terminal summary or route plot in the box below.

Baseline screenshot / plot

Paste a screenshot of the modified terminal summary or route plot in the box below.

Modified-case screenshot / plot

8. Develop recommendations

Based on the comparison, recommend two changes that could improve evacuation readiness or robustness. Recommendations may involve mobile charging resources, pre-evacuation charging/range, traffic management, or demand management.

Proposed improvement	Supporting result	Implementation constraint / information needed

9. Prepare your group report

Question	Group response
What parameter did your group change?	
What was your most important finding?	
Which OD pair was most sensitive to the change?	
How did MCS deployment change?	
What limitation of this analysis should be considered when interpreting the result?	
What additional local data would improve the analysis?	

10. Instructor / participant notes

Template structure adapted from the uploaded participant worksheet format; optimization inputs and outputs are tailored to the EV evacuation routing/MCS code.